# US Patent & Trademark Office
# Patent Public Search | Text View


| | |
|---|---|
| United States Patent Application Publication | 20250285778 |
| Kind Code | A1 |
| Publication Date | September 11, 2025 |
| Inventor(s) | Brunetti; Alessandro et al. |


## IRRADIATION STATION FOR A RADIOISOTOPE PRODUCTION SYSTEM

### Abstract


An irradiation station for producing a radioisotope, having a cyclotron for emitting a proton beam against a solid target material placed in a container and a cooling system for cooling the container. The container has a wall for supporting the solid target material and a cavity, which borders the wall and has an opening transverse to the axis of the container. The cooling system has a connection head which is couplable to the opening for circulating a cooling fluid in the cavity and comprises a flow diverter, a tip protruding from a hole of the flow diverter and movable along the hole against the action of a spring in contact with the tip, and an electrical connector in contact with the spring. When the connection head is coupled to the opening the flow diverter enters the cavity coaxially with the container and the tip presses against the wall.



| | |
|---|---|
| **Inventors:** | **Brunetti; Alessandro (Castel Bolognese (RA), IT), Testa; Marco (Castel Bolognese (RA), IT)** |
| **Applicant:** | **COMECER S.P.A.** (Castel Bolognese (RA), IT) |
| **Family ID:** | **82781342** |
| **Appl. No.:** | **18/859384** |
| **Filed (or PCT Filed):** | **April 24, 2023** |
| **PCT No.:** | **PCT/IB2023/054186** |

### Foreign Application Priority Data

| | | |
|---|---|---|
| IT | 102022000008456 | Apr. 28, 2022 |

### Publication Classification

**Int. Cl.:** **G21G1/10** (20060101)

**U.S. Cl.:**

CPC **G21G1/10** (20130101);

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This Patent Application claims priority from Italian Patent Application No. 102022000008456 filed on Apr. 28, 2022, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to an irradiation station for a radioisotope production system.

[0003] In particular, the invention is advantageously, but not exclusively, applied to the production of a radioisotope utilizing a medium- or high-energy cyclotron, i.e. a cyclotron with energy equal to or above 18 MeV, starting from a solid precursor material, also known as solid target material, in the form of a thin layer electrodeposited on a suitable metal support, or of a foil positioned on the metal support, or of a capsule of compressed powder positioned on the metal support, to which the following description will explicitly refer without thereby losing generality.

BACKGROUND

[0004] To date, various types of radioisotopes for pharmaceutical use (radiopharmaceuticals) are obtained following the irradiation by means of a beam of protons (proton bombardment) of a solid target material typically having a metal origin.

[0005] The production process of a radioisotope starting from a solid target material substantially provides for the following steps: positioning a portion of solid target material on a metal support; irradiating he solid target material on the support by means of proton beam; dissolving the irradiated solid target material for obtaining a solution in which the radioisotope produced by the proton irradiation is present; and purifying the aforementioned solution for separating the radioisotope from the target material which did not react and from impurities. The positioning of the solid target material on the metal support takes place according to modes that depend on the type and form of solid target material utilized. For example, the solid target material is in the form of a thin portion electrodeposited on the support, or of a metal foil or of a capsule of compressed powder.

[0006] The aforementioned steps are carried out in relative processing stations and therefore the support carrying the solid target material has to be arranged inside a container in order to be transported between various processing stations, for example from the electrodeposition station to the irradiation station and from the irradiation station to the dissolution station.

[0007] Radioisotope production systems are known which comprise a positioning station, an irradiation station, a dissolution station, a purification station and an automated transportation system for the transportation, between some of the aforementioned stations, of the container containing the support with the solid target material still to be irradiated or already irradiated. For such reason, such container is also known as shuttle.

[0008] The irradiation station comprises a cyclotron for emitting the proton beam against the solid target material and a fluid cooling system which is connected to the support for the relative cooling during the proton bombardment. Furthermore, supports are known designed to be placed directly in the dissolution station and capable of resisting to the agents which produce the solution with the radioisotope.

[0009] The main factors that affect the productivity (mCi/uAh) of a radioisotope production system are the parameters of the proton beam, geometry and materials of the solid target material

container, the type and the geometry of the solid target material, and the degradation of the beam due to possible obstacles which the proton beam passes through before reaching the solid target material. The main parameters of the proton beam which have an impact on the productivity are beam current (uA), beam energy (MeV) and beam diameter (mm).
[0010] For example, the type and geometry of the solid target material being equal, the increase in the beam current allows increasing the productivity. other In words, other parameters being equal, the productivity is directly linked to the beam current. However, the beam current set on the cyclotron may not result in the desired productivity because of the variability of the multiple parameters affecting it.
[0011] Furthermore, the increase in the beam current causes an increase in the temperature of the irradiation station and of the solid target material which has to be compensated by the fluid cooling system. In particular, in order not to compromise the operation of the irradiation station, the temperature thereof and that of the solid target material has to be kept within the range comprised between 150 and 200° C. Should the temperature exceed the aforementioned range, the different thermal expansion between the material of the metal support of the container and the solid target material could cause the disjunction of the latter from the metal support with consequent production stop and therefore a drop in the radioisotope production.
SUMMARY
[0012] The object of the present invention is to provide an irradiation station which is exempt from the drawbacks described above and, simultaneously, is easy and cost-effective to manufacture.
[0013] In accordance with the present invention, an irradiation station for a radioisotope production system and a radioisotope production system are provided according to what is defined in the appended claims.

---

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] The present invention will now be described with reference to the accompanying drawings, which illustrate a non-limiting example embodiment thereof, wherein:
[0015] FIG. **1** illustrates an exploded axonometric view of a container for a solid target material;
[0016] FIG. **2** illustrates the container of FIG. **1** according to a section view along a plane on which the longitudinal axis of the container **1** lies;
[0017] FIGS. **3** and **4** illustrate a detail of FIG. **2** during two different uses of the container;
[0018] FIG. **5** illustrates, in part and according to a section view, the irradiation station of the present invention for a radioisotope production system which utilizes the container of FIG. **1**; and
[0019] FIG. **6** illustrates an axonometric view of a part of a cooling system of the irradiation station of FIG. **1**.
DESCRIPTION OF EMBODIMENTS
[0020] In FIGS. **1** and **2**, reference numeral **1** generically indicates, as a whole, the container of the present invention suitable to contain a solid target material and a radioisotope produced by means of irradiation with proton beam of the solid target material.
[0021] The container **1** extends according to a longitudinal axis **2** thereof and comprises a well-shaped body **3**, in the following simply called well, which is designed to support, on a side **4***a* of a bottom wall **4** thereof, a portion of solid target material (not illustrated), a support body **5**, which extends according to the longitudinal axis **2** and comprises a first portion **6** having a seat **7** designed, in use, to coaxially house the well **3** in such a way that the bottom wall **4** is arranged transverse to the longitudinal axis **2**, and a lid **8**, which comprises a cup-shaped central portion **9**, the bottom of which comprises a degrading foil **10** designed to mitigate the proton beam in a pre-established manner. The lid **8** is designed, in use, to be coaxially fitted on the portion **6** in such a

way that the central portion **9** is arranged in the well **3** for holding the portion of solid target material on the bottom wall **4** and that the degrading foil **10** is arranged above, and in particular parallel to, the bottom wall **4** in such a way that the portion of solid target material is arranged, in use, between the degrading foil **10** and the bottom wall **4**.

[0022] In use, a proton beam (not illustrated in FIGS. **1** and **2**) is directed on the central portion **9** in the direction of the longitudinal axis **2**, and in particular centred on the longitudinal axis **2**, for striking the degrading foil **10** in a substantially perpendicular manner. The degrading foil **10** has a thickness calibrated for mitigating the proton beam in such a measure to transfer to the portion of solid target material arranged in the well **3** a medium energy (MeV) which allows obtaining the desired radioisotope. For example, the thickness of the degrading foil **10** is comprised between 50 μm and 500 μm. The value of the thickness is chosen depending on the radioisotope to be produced. In particular, each radioisotope to be produced is associated with a respective lid **8** having a degrading foil **10** with a specific thickness which is sized depending on the radioisotope.

[0023] The support body **5** has a shape of cylindrical symmetry with respect to the longitudinal axis **2**. The well **3** has a shape of cylindrical cup, i.e. a cylinder devoid of a base. Also the lid **8** has a shape of cylindrical symmetry.

[0024] The support body **5** comprises a second portion **11**, which is coaxial to the portion **6**. The portion **11** comprises an internal cavity **12**, which communicates with the seat **7** through an opening **13** coaxial to the longitudinal axis **2** and with the outside through a further opening **14** (FIG. **2**) transverse to the longitudinal axis **2** for allowing the access of a cooling fluid in the cavity **12**. As is evident in FIG. **2**, the bottom wall **4** of the well **3** closes the opening **13** when the well **3** is in the seat **7** in such a way that a side **4***b* of the bottom wall **4** opposite the side **4***a*, in use, is lapped by the cooling fluid. The cavity **12** has a shape of cylindrical symmetry with respect to the longitudinal axis **2** and therefore also the portion **11** has a similar cylindrical shape.

[0025] The seat **7** houses the well **3** with hermetic interference between a side internal surface **15** (FIG. **1**) of the seat **7** and a side external surface **16** (FIG. **1**) of the well **3**. Such hermetic interference is obtained with a precise machining of the side internal surface **15** and of the side external surface **16**. The side hermetic interference between seat **7** and well **3** prevents, in use, the cooling fluid from passing through the opening **13** and ending up in the well **3**. The portion **6** of the support body **5** comprises an external thread **17** and the lid **8** comprises an annular portion **18**, which is arranged around, in a coaxial manner, the central portion **9** and comprises an internal thread **19** (FIG. **2**) for being screwed to the portion **6**.

[0026] The container **1** further comprises a hermetic sealing ring **20**, which is fitted on the support body **5**. In particular, the hermetic sealing ring **20** is held in a groove **21** of the support body **5** arranged between the portion **6** and the portion **11**. With particular reference to the enlarged detail of FIG. **2**, the hermetic sealing ring **20** enters into contact with the support body **5**, and in particular with the groove **21**, and an internal surface **22** of an end portion **23** of the annular portion **18** of the lid **8** when the annular portion **18** is screwed to the portion **6**. In this manner, the lid **8** hermetically seals the well **3** for preventing radioactive substances from coming out of the well **3** during the production of the radioisotope.

[0027] The lid **8** comprises a plurality of external notches **24** and similarly the portion **11** of the support body **5** comprises a plurality of external notches **25** for facilitating the gripping of the fingers of an operator during the closing of the container **1**. In particular, the notches **25** are arranged along an end portion **26** of the portion **11** surrounding the opening **14**. The end portion **26** is substantially cylindrical with respect to the longitudinal axis **2**. The end portion **26** ends with a substantially axial circular edge **26***a* surrounding the opening **14**.

[0028] The support body **4** and the lid **8** are made of aluminium, which is an easily workable metal. The well is made of a material suitable to the electrodeposition of the solid target material and is inert to acid substances capable of dissolving the portion of solid target material. Preferably, the well **3** is internally made of platinum. Advantageously, all the walls of the well **3** have a thickness less than 1 mm, in particular approximately 500 μm.

[0029] With particular reference to FIG. **2**, the central portion **9** comprises an annular rib **27** surrounding the degrading foil **10** and projects from the plane of the degrading foil **10** parallel to the longitudinal axis **2** so as to end with an end surface **28**, it too annular, designed to press against the bottom wall **4** of the well **3** when the lid **8** is fitted on the portion **6** so as to define, between the degrading foil **10** and the bottom wall **4**, a chamber **29** centred on the longitudinal axis **2** for containing a portion of solid target material. The structure of the central portion **9** allows containing the solid target material in various sizes.

[0030] FIG. **3** more specifically illustrates a portion of the container **1** around the chamber **29** in a utilization example in which the portion of solid target material is in the form of metal foil, indicated by M**1**, which is spread on the bottom of the well **3**, i.e. on the bottom wall **4**. In use, the lid **8** is fitted on the portion **6** and the annular portion **18** is screwed to the portion **6** until the end surface **28** of the rib **27** presses an edge of the metal foil M**1** against the bottom wall **4**. The portion of metal foil M**1** facing within the chamber **29** will be the one irradiated by the proton beam passing through the degrading foil **10**.

[0031] FIG. **4** illustrates the same portion of the container **1** of FIG. **3** in a different utilization example in which the portion of solid target material is in the form of a capsule of compressed powder, indicated by M**2**, which is housed in the chamber **29**. In use, the lid **8** is fitted on the portion **6** and the annular portion **18** is screwed on the portion **6** until the end surface **28** of the rib **27** enters into contact with the bottom wall **4**. In this manner, the capsule of compressed power M**2** is held by the chamber **29** and in centred position on the longitudinal axis **2**. The capsule of compressed powder M**2** will thus be completely irradiated by the proton beam through the degrading foil **10**.

[0032] In a further utilization example not illustrated, the portion of solid target material is in the form of a thin layer of material electrodeposited on the bottom wall **4** of the well **3** so as to remain within the chamber **29**, i.e. completely underneath the degrading foil **10** so as to be irradiated by the proton beam which strikes the degrading foil **10**.

[0033] FIG. **5** illustrates in a simplified manner a part of an irradiation station **30** of a radioisotope production system which comprises the container **1**. The radioisotope production system is in general known per se, and thus not specifically illustrated. The container **1** is used in the radioisotope production system for transferring the portion of solid target material between several stations of the system, among which the irradiation station **30**. FIG. **5** illustrates the irradiation station **30** according to a section view along a plane on which an alignment axis **31** of the irradiation station **30** lies.

[0034] The irradiation station **30** comprises a cyclotron **32** of known type for emitting the proton beam B against the portion of solid target material, for example in the form of a thin layer of electrodeposited material indicated by M, arranged in the container **1**. In particular, the cyclotron **32** is of the type capable of emitting a proton beam with energy equal to or above **18** MeV. The proton beam B passes through the degrading foil **10**, which offers a pre-established mitigation, and irradiates the portion of solid target material which is in the well **3** laid on the bottom wall **4**.

[0035] The irradiation station **30** comprises a fluid cooling system **33** connectable to the container **1** for cooling the latter during the irradiation with proton beam B of the solid target material. In particular, the fluid cooling system **33** comprises a connection head **34** designed to couple to the opening **14** of the support body **5** so as to circulate a cooling fluid in the cavity **12**.

[0036] The container **1** is illustrated in FIG. **5** according to the same section view of FIG. **2**. The connection head **34** comprises a delivery conduit **35** and a return conduit **36** for a cooling fluid which face the opening **14** and a flow diverter **37**, which is in part arranged between the delivery conduit **35** and the return conduit **36**, extends along the axis **31** and is designed to enter the cavity **12** through the opening **14** coaxially to the container **1**, i.e. with the axis **31** coinciding with the longitudinal axis **2**.

[0037] In particular, the connection head **34** comprises a body **38**, which is made of aluminium and comprises the flow diverter **37** and in which the delivery conduit **35** and the return conduit **36**, and a centring ring nut **39** are obtained, the latter has a first portion **40** screwed to an external threaded portion of the body **38** and a second portion **41** defining a cylindrical seat for the end portion **26** of the container **1**. Therefore, the centring ring nut **39** supports the container **1** keeping it centred on the connection head **34**, and in particular coaxial to the flow diverter **37**.
[0038] The centring ring nut **39** also determines the positioning of the container **1** with respect to the cyclotron **32** and thus with respect to the proton beam B. In fact, the cyclotron **32** is aligned with the connection head **34** in such a way that the central portion **9** of the lid **8** is faced towards the cyclotron **32** and the proton beam B is directed on the degrading foil **10** parallel to the longitudinal axis **2**, and in particular is centred on the degrading foil **10**.
[0039] The flow diverter **37** is shaped so as to define in the cavity **12** a circulation path **42** for the cooling fluid. In particular, the circulation path **42** is U-shaped and extends from the delivery conduit **35** to the return conduit **36**. More specifically, the circulation path **42** comprises an input section **43** and an output section **44** for the cooling fluid which are parallel to the longitudinal axis **2**. The input section **43** communicates with the delivery conduit **35** and the output section **44** communicates with the return conduit **36**.
[0040] At the opening **13**, and in particular parallel to the opening **13**, the circulation path **42** comprises an intermediate section **45** transverse to the longitudinal axis **2** in such a way that, in use, the cooling fluid assumes a laminar flow at least along the intermediate section **45**. The laminar flow will lap the side **4***b* of the bottom wall **4** of the well **3** facing from the opening **13**.
[0041] The flow diverter **37** comprises a hole **46** centred on the axis **31** and the connection head **34** comprises a tip **47**, which protrudes, in part, from the hole **46** and is movable along the hole **46** against the action of a spring **48**. The spring is housed in the body **38** so as to be centred on the axis **31** and is in electrical contact with the tip **47**. Preferably, the spring **48** is a helical spring. In particular, the flow diverter **37** comprises a further hole **49**, which is coaxial with the hole **46** and communicates with the hole **46** for housing the spring **48**.
[0042] Preferably, the hole **49** has a greater diameter than the hole **46** and the tip **47** has an enlarged end **50**, which slides in the hole **49** and is in contact with a first end of the spring **48**. In the absence of the container **1**, i.e. in a configuration not illustrated in the figures, the enlarged end **50**, thanks to the thrust of the spring **48**, abuts against a shoulder **51** defined by the different diameters of the holes **46** and **49** in such a way that the tip **47** protrudes from the hole **46** with a maximum extension.
[0043] When the connection head **34** is coupled to the opening **14**, as is illustrated in FIG. **5**, the tip **48** passes through the opening **13** and presses against the side **4***b* of the bottom wall **4** of the well **3**, partially re-entering the hole **46**. The spring **48** compresses and keeps the tip **47** pressed against the side **4***b* of the bottom wall **4** thus creating an electrical contact between the tip **47** and the bottom wall **4**.
[0044] The connection head **34** comprises an electrical connector **52** which is in electrical contact with the spring **48**, in particular with the second end of the spring **48**. Specifically, the electrical connector **52** comprises a first portion **53**, which hydraulically seals an end of the hole **49** and has a pin **54** jutting in the hole **49** and engaging with contact the second end of the spring **48**, and a second portion **55**, which comprises an electrical socket **56** of the jack type accessible from the outside. An O-ring **57** fitted on the portion **53** ensures the hydraulic sealing between the portion **53** and an enlarged portion of the hole **49**.
[0045] A current meter (not illustrated), which is optionally part of the irradiation station **30**, is connectable to the electrical connector **52**, and in particular to the electrical socket **56** by means of a plug of the jack type, for measuring an electric current in the portion of solid target material M which is induced by the proton beam B and which is thus an indirect indication of the beam current set on the cyclotron **32**.

[0046] Preferably, the tip **47** is made of copper for reducing the electric resistance thereof and is gold-plated for improving the electrical contact with the bottom wall **4** of the well **3** of the container **1**. Preferably, the spring **48** is made of stainless steel and is gold-plated for improving the electrical contact with the tip **47** and with the pin **54**. Preferably, the electrical connector **52** is made of copper for reducing the electric resistance thereof and is gold-plated for improving the electrical contact with the spring **48**.

[0047] The connection head **34** comprises an annular seal **58**, which is arranged in an axial annular seat **59** of the body **38** and has a flat face suitable for sealing the edge **26***a* of the end portion **26** of the container **1**.

[0048] With reference also to FIG. **6**, which illustrates the body **38** according to a front axonometric view, the flow diverter **37** has a transverse dimension which is predominant along a direction **31***a* orthogonal to the axis **31** and the **36** a delivery conduit **35** and the return conduit have transverse section which is flattened along a direction **31***b* orthogonal to the axis **31** and to the direction **31***a* for favouring, in use, i.e. inside the cavity **12** of the container **1**, the formation of the laminar flow at a top **60** of the flow diverter **37**. The hole **46** from which the tip **47** protrudes opens at the top **60**. The delivery conduit **35** and the return conduit **36** extend inside the body **38** in the form of respective fitting holes **35***a* and **36***a*, which open at a rear portion **61** of the body **38** and are preferably parallel to the axis **31**.

[0049] Still with reference to FIG. **5**, the connection head **34** comprises a further body **62**, which is made of PEEK, i.e. polyether ether ketone, comprises two through holes **63**, and is hydraulically coupled to the body **38** in such a way that a first one of the through holes **63** communicates with the delivery conduit **35**, in particular through the fitting hole **35***a*, and the second one of the through holes **63** communicates with the return conduit **36**, in particular through the fitting hole **36***a*.

[0050] PEEK is a material that insulates from the electric currents. In such manner, the connection head **34** with the body **62** made of PEEK allows electrically insulating a part of the cooling system **33** from the cyclotron **32**.

[0051] The hydraulic sealing between the body **38** and the body **62** is ensured by three O-rings. In particular, one O-ring **64** is fitted on the rear portion **61** and the latter engages a radial seat **65** of the body **62**. Furthermore, two further O-rings **66** are arranged in respective seats of a face of the rear portion **61** transverse to the axis **31** for each ensuring the hydraulic sealing between a respective through hole **63** and a respective fitting hole **35***a*, **36***a*.

[0052] Each through hole **63** is provided with a respective fitting **67** to which a respective pipe **68** is connected in which the cooling fluid circulates. Each fitting **67** is, in turn, provided with a respective adjuster **69** for the pipe **68**.

[0053] The body **62** has a further through hole **70** passed through by the portion **55** of the electrical connector **52**.

[0054] An advantage of the irradiation station **30** described above is a better cooling of the well **3** and thus of the portion of solid target material during the irradiation step of the latter, thanks to the laminar flow of the cooling fluid which laps the side **4***b* of the bottom wall **4** through the opening **13** of the internal cavity **12** of the support body **5**. In particular, the flow diverter **37** entering the cavity **12** for defining therein a U-shaped circulation path **36**, allows the cooling fluid to assume, at the opening **13**, and in particular in the intermediate section **45** of the circulation path **42**, a laminar flow which laps the side **4***b* of the bottom wall **4**.

[0055] Another advantage of the irradiation station **30** is the possibility to measure the current circulating in the solid target material while it is irradiated by the proton beam, thanks to the presence of the tip **47** protruding from the hole **46** at the top **60** of the flow diverter **37** and which enters into contact with the side **4***b* of the bottom wall **4** of the well **3** by effect of the thrust of the spring **48**. Furthermore, the tapered shape of the tip **47** does not bother the laminar flow of the cooling fluid which is formed in the intermediate section **45** of the circulation path **42**, i.e. between the top **60** of the flow diverter **37** and the bottom wall **4**.

## Claims

**1**. An irradiation station for a radioisotope production system, the irradiation station (**30**) comprising a cyclotron (**32**) for emitting a proton beam (B) against a portion of solid target material (M) contained in a container (**1**) and a cooling system (**33**) for cooling the container (**1**) during the irradiation of the portion of solid target material (M); the container (**1**) extending along a first axis (**2**) and comprising a support wall (**4**) for supporting, on a first side (**4***a*), the portion of solid target material (M) and an internal cavity (**12**), which borders the support wall (**4**), on a second side (**4***b*) opposite the first side (**4***a*), and communicates with the outside through a first opening (**14**) transverse to the first axis (**2**); the cooling system (**33**) comprising a connection head (**34**) and the latter comprises: a delivery conduit (**35**) and a return conduit (**36**) for a cooling fluid; a flow diverter (**37**) extending along a second axis (**31**); a tip (**47**), which protrudes, in part, from a hole (**46**) of the flow diverter (**37**) and is movable along said hole (**46**) against the action of a spring (**48**) housed in the connection head (**34**) and in electrical contact with the tip (**47**), said hole (**46**) being centred on the second axis (**31**); and an electrical connector (**52**) in electrical contact with the spring (**48**); the connection head (**34**) being designed to couple to the first opening (**14**) for circulating the cooling fluid in the cavity (**12**) and in such a way that the flow diverter (**37**) enters the cavity (**12**) coaxially with the container (**1**) and the tip (**47**) presses against said second side (**4***b*).
**2**. The irradiation station according to claim 1, and comprising a current meter connected to the electrical connector (**52**) for measuring an electric current in the portion of solid target material (M) induced by the proton beam (B).
**3**. The irradiation station according to claim 1, wherein said flow diverter (**37**) is shaped to define in the cavity (**12**) a U-shaped circulation path (**42**) for the cooling fluid extending from the delivery conduit (**35**) to the return conduit (**36**).
**4**. The irradiation station according to claim 3, wherein said circulation path (**42**) comprises an intermediate section (**45**) transverse to the first axis (**2**) so that, in use, the cooling fluid assumes a laminar flow at least along the intermediate section (**45**).
**5**. The irradiation station according to claim 1, wherein said spring (**48**) is helical, is arranged in a housing (**49**) of the connection head (**34**) and has a first end in contact with a portion (**50**) of the tip (**47**) which is movable within the housing (**49**); said electrical connector (**52**) comprising a pin (**54**), which engages with contact a second end of the spring (**48**), and an electrical socket (**56**) of the jack type accessible from the outside.
**6**. The irradiation station according to claim 1, wherein said container (**1**) comprises a cylindrical portion (**26**) ending with an edge (**26***a*) surrounding the first opening (**14**) and said connection head (**34**) comprises a first body (**38**), which is made of aluminium and comprises said flow diverter (**37**), delivery conduit (**35**) and return conduit (**36**), and an annular seal (**58**), which is arranged in an annular seat (**59**) of the first body (**38**) and has a flat face suitable for sealing said edge (**26***a*).
**7**. The irradiation station according to claim 1, wherein said connection head (**34**) comprises a first body (**38**), which is made of aluminium and comprises said flow diverter (**37**), delivery conduit (**35**) and return conduit (**36**), and a second body (**62**), which is made of PEEK, comprises two through-holes (**63**), each provided with a respective fitting (**67**) for a respective pipe (**68**) for the cooling fluid, and is hydraulically coupled to the first body (**38**) in such a way that a first through-hole (**63**) communicates with the delivery conduit (**35**) and the second through-hole (**63**) communicates with the return conduit (**36**).
**8**. The irradiation station according to claim 1, wherein said container (**1**) comprises a cylindrical portion (**26**) surrounding the first opening (**14**) and said connection head (**34**) comprises a first body (**38**), which is made of aluminium and comprises said flow diverter (**37**), delivery conduit (**35**) and return conduit (**36**), and a centring ring nut (**39**), which has a first portion (**40**) screwed to

the outside of the first body (**38**) and a second portion (**41**) defining a cylindrical seat for the cylindrical portion (**26**) of the container (**1**).

**9**. A radioisotope production system comprising a container (**1**) for containing a portion of solid target material (M) and an irradiation station (**30**); the container (**1**) extending along a first axis (**2**) and comprising a support wall (**4**) for supporting, on a first side (**4***a*), the portion of solid target material (M) and an internal cavity (**12**), which borders the support wall (**4**), on a second side (**4***b*) opposite the first side (**4***a*), and communicates with the outside through a first opening (**14**) transverse to the first axis (**2**); the irradiation station (**30**) being according to claim 1.

**10**. A radioisotope production system according to claim 9, wherein said container (**1**) comprises a well (**3**), which has a bottom wall coinciding with said support wall (**4**), and a support body (**5**) comprising a first portion (**6**), which has a seat (**7**) for housing the well (**3**), and a second portion (**11**), which has said internal cavity (**12**) and communicates with the seat (**7**) through a second opening (**13**) transverse to said first axis (**2**) in such a way that, in use, when said connection head (**34**) is coupled to said first opening (**14**), the tip (**47**) passes through said second opening (**13**).